variable declaration in python

In [1]:
```
# Python variable declaration and assign value (name=value)
i = 1030
i
```

Out[1]:
```
1030
```

In [1]:
```
# Python variable name should not start with a number
1030 = i
```

```
  Cell In[1], line 2
    1030 = i
    ^
SyntaxError: cannot assign to literal here. Maybe you meant '==' instead of '='?
```

In [5]:
```
# Python variable name should not start with a number
1nit = 10
1nit
```

```
  Cell In[5], line 1
    1nit = 10
    ^
SyntaxError: invalid decimal literal
```

In [7]:
```
#Python variable name can have number at the end
nit1 = 100
nit1
```

Out[7]:
```
100
```

In [8]:
```
# Python variable names are case sensitive
nit = 20
NIT
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[8], line 2
      1 nit = 20
----> 2 NIT

NameError: name 'NIT' is not defined
```

In [10]:
```
nit
```

Out[10]:
```
20
```

In [9]:
```
# Python variable name should not contain special characters other than _
nit@ = 8
nit@
```

```
  Cell In[9], line 2
    nit@ = 8
         ^
SyntaxError: invalid syntax
```

In [11]:
```
# Python variable name should not contain special characters other than _
nit_= 8
nit_
```

Out[11]: 8

In [12]:
```
# Python variable name should not contain special characters other than _

_nit= 8
_nit
```

Out[12]: 8

In [14]:
```
# Python keyword list

import keyword
keyword.kwlist
```

Out[14]:
```
['False',
 'None',
 'True',
 'and',
 'as',
 'assert',
 'async',
 'await',
 'break',
 'class',
 'continue',
 'def',
 'del',
 'elif',
 'else',
 'except',
 'finally',
 'for',
 'from',
 'global',
 'if',
 'import',
 'in',
 'is',
 'lambda',
 'nonlocal',
 'not',
 'or',
 'pass',
 'raise',
 'return',
 'try',
 'while',
 'with',
 'yield']
```

In [15]:
```
len(keyword.kwlist)
```

Out[15]:
```
35
```

In [16]:
```
#pyhthon variable name should not use keywords

for = 45
for
```

```
  Cell In[16], line 1
    for = 45
        ^
SyntaxError: invalid syntax
```

In [17]:
```
For = 50
For
```

Out[17]:
```
50
```

In [18]:
```
if = 67
```

```
  Cell In[18], line 1
    if = 67
       ^
SyntaxError: invalid syntax
```

In [20]:
```
If = 67
If
```

Out[20]: 67

In [21]:
```
else = 7
```

```
  Cell In[21], line 1
    else = 7
    ^
SyntaxError: invalid syntax
```

10th--python datatypes

In [23]:
```
i = 5
i
```

Out[23]: 5

In [24]:
```
type()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[24], line 1
----> 1 type()

TypeError: type() takes 1 or 3 arguments
```

In [25]:
```
# integer datatype

type(i)
```

Out[25]: int

In [26]:
```
id(i)
```

Out[26]: 140710759736360

In [27]:
```
# float datatype

f = 110.45
f
```

Out[27]: 110.45

In [28]:
```
type(f)
```

Out[28]: float

In [29]:
```
gold_price = 14800.78
gold_price
```

Out[29]: 14800.78

In [30]:
```
type(gold_price)
```

Out[30]: float

In [31]:
```
10 + 5
10 - 5
10 * 5
```

Out[31]: 50

In [32]:
```
# print datatype

print(10+5)
print(10-5)
print(10*5)
```

```
15
5
50
```

In [35]:
```
# float datatype with exponential

f1 = 2.4e1
f1
```

Out[35]: 24.0

In [34]:
```
type(f1)
```

Out[34]: float

In [36]:
```
f2 = 3.5e3
f2
```

Out[36]: 3500.0

boolean datatype

In [37]:
```
true
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[37], line 1
----> 1 true

NameError: name 'true' is not defined
```

In [38]:
```
True
```

Out[38]:
```
True
```

In [39]:
```
false
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[39], line 1
----> 1 false

NameError: name 'false' is not defined
```

In [40]:
```
False
```

Out[40]:
```
False
```

In [41]:
```
True + True
```

Out[41]:
```
2
```

In [42]:
```
True + False - False
```

Out[42]:
```
1
```

In [43]:
```
True * True
```

Out[43]:
```
1
```

In [44]:
```
# float division
False / True
```

Out[44]:
```
0.0
```

In [45]:
```
False// True
```

Out[45]:
```
0
```

In [46]:
```
True / False
```

```
---------------------------------------------------------------------------
ZeroDivisionError                         Traceback (most recent call last)
Cell In[46], line 1
----> 1 True / False

ZeroDivisionError: division by zero
```

In [47]:
```
int(True)
```
Out[47]: 1

complex datatype

In [48]:
```
c = 10 + 20j # 10 - real part & 20 imaginary part where j = squre root of -1
c
```
Out[48]: (10+20j)

In [49]:
```
type(c)
```
Out[49]: complex

In [50]:
```
c = 10 + 20j
c
```
Out[50]: (10+20j)

In [51]:
```
c.real
```
Out[51]: 10.0

In [53]:
```
c.imag
```
Out[53]: 20.0

11th

integer, float, complex, sting, bool

In [1]:
```
a=4
a
```
Out[1]: 4

In [2]:
```
type(a)
```
Out[2]: int

In [3]:
```
b=5.9
b
```
Out[3]: 5.9

In [4]:
```
Type(b)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[4], line 1
----> 1 Type(b)

NameError: name 'Type' is not defined
```

In [5]:
```
type(b)
```

Out[5]:
```
float
```

In [8]:
```
a=True
a1=False
b=a+a1
b
```

Out[8]:
```
1
```

In [9]:
```
type(b)
```

Out[9]:
```
int
```

In [10]:
```
True + False
```

Out[10]:
```
1
```

In [11]:
```
b = False
type(b)
```

Out[11]:
```
bool
```

In [12]:
```
k=true
type(k)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[12], line 1
----> 1 k=true
      2 type(k)

NameError: name 'true' is not defined
```

In [13]:
```
k=True
type(k)
```

Out[13]:
```
bool
```

In [14]:
```
c = 5+6j
c
```

Out[14]:
```
(5+6j)
```

In [15]:
```
type(c)
```

Out[15]:
```
complex
```

In [16]:
```
c
```

Out[16]:
```
(5+6j)
```

In [17]:
```
c.real
```

Out[17]:
```
5.0
```

In [20]:
```
c.imaginary
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[20], line 1
----> 1 c.imaginary

AttributeError: 'complex' object has no attribute 'imaginary'
```

In [21]:
```
c.imag
```

Out[21]:
```
6.0
```

In [22]:
```
c
```

Out[22]:
```
(5+6j)
```

In [23]:
```
h= 2+3j
h
```

Out[23]:
```
(2+3j)
```

In [24]:
```
c+h
```

Out[24]:
```
(7+9j)
```

In [25]:
```
print(c)
print(h)
```

```
(5+6j)
(2+3j)
```

In [26]:
```
print(c+h)
```

```
(7+9j)
```

string

In [30]:
```
welcome to nareshit & request everyone please subscr nareshit youtube, instagram
```

```
  Cell In[30], line 1
    welcome to nareshit & request everyone please subscr nareshit youtube, instagram
            ^
SyntaxError: invalid syntax
```

In [31]:
```
'welcome to nareshit & request everyone please subscr nareshit youtube, instagram'
```

Out[31]:
```
'welcome to nareshit & request everyone please subscr nareshit youtube, instagram'
```

In [32]:
```
"welcome to nareshit & request everyone please subscr nareshit youtube, instagram"
```

Out[32]:
```
'welcome to nareshit & request everyone please subscr nareshit youtube, instagram'
```

In [33]:
```
'''welcome to nareshit & request everyone please subscr nareshit youtube, instagram
```

Out[33]:
```
'welcome to nareshit & request everyone please subscr nareshit youtube, instagram'
```

In [34]:
```
''''welcome to nareshit & request everyone please subscr nareshit youtube, instagra
```

```
  Cell In[34], line 1
    ''''welcome to nareshit & request everyone please subscr nareshit youtube, insta
gram''''

^
SyntaxError: unterminated string literal (detected at line 1)
```

In [35]:
```
'welcome to nareshit
request everyone please
subscr nareshit youtube, instagram'
```

```
  Cell In[35], line 1
    'welcome to nareshit
    ^
SyntaxError: unterminated string literal (detected at line 1)
```

In [36]:
```
"welcome to nareshit
request everyone please
subscr nareshit youtube, instagram"
```

```
  Cell In[36], line 1
    "welcome to nareshit
    ^
SyntaxError: unterminated string literal (detected at line 1)
```

In [37]:
```
'''welcome to nareshit
request everyone please
subscr nareshit youtube, instagram'''
```

Out[37]:
```
'welcome to nareshit \nrequest everyone please \nsubscr nareshit youtube, instagra
m'
```

string indexing

string forward indexing

In [38]:
```
s = 'python'
s
```

Out[38]: 'python'

In [39]:
```
s1 = 'welcome'
s1
```

Out[39]: 'welcome'

In [43]:
```
s1[3]
```

Out[43]: 'c'

In [44]:
```
s1[6]
```

Out[44]: 'e'

In [45]:
```
s1[9]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[45], line 1
----> 1 s1[9]

IndexError: string index out of range
```

string backward indexing

In [46]:
```
s1
```

Out[46]: 'welcome'

In [48]:
```
s1[-2]
```

Out[48]: 'm'

In [49]:
```
s1[-5]
```

Out[49]: 'l'

In [50]:
```
s1[-8]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[50], line 1
----> 1 s1[-8]

IndexError: string index out of range
```

In [51]:
```
len(s1)
```

Out[51]:
```
7
```

In [52]:
```
s1[0]
```

Out[52]:
```
'w'
```

In [53]:
```
s1
```

Out[53]:
```
'welcome'
```

forward slicing

In [54]:
```
s1[:]
```

Out[54]:
```
'welcome'
```

In [55]:
```
s1[3:]
```

Out[55]:
```
'come'
```

In [56]:
```
s1[:6]
```

Out[56]:
```
'welcom'
```

In [57]:
```
s1[2:5]
```

Out[57]:
```
'lco'
```

In [59]:
```
name= 'mynameisniharika'
name
```

Out[59]:
```
'mynameisniharika'
```

In [60]:
```
name[:]
```

Out[60]:
```
'mynameisniharika'
```

In [61]:
```
len(name)
```

Out[61]:
```
16
```

In [63]:
```
name[6:16]
```
Out[63]:
```
'isniharika'
```
In [64]:
```
name[:15]
```
Out[64]:
```
'mynameisniharik'
```
In [65]:
```
name[4:]
```
Out[65]:
```
'meisniharika'
```
In [66]:
```
name[3:10:3]
```
Out[66]:
```
'aii'
```
In [ ]:
```
```